

Technical Datasheet 4G Gateway with 4DI, 4DO, 2AI, 1PT, 1TH

Product Overview: -

The 4G Gateway is an advanced networking device designed to provide seamless connectivity and data management capabilities for various applications. With support for 4G LTE connectivity, cloud connectivity via MQTT protocol, and robust storage options, it offers a comprehensive solution for remote monitoring and control systems. Equipped with Modbus RS485 port and micro Sim compatibility, it caters to a wide range of industrial and IoT applications. The intuitive web-based UI interface allows for easy configuration and management, making it an ideal choice for both professionals and enthusiasts.



Specifications

General –

- **4G LTE Connectivity:** Utilizes 4G LTE technology for reliable and high-speed internet access. If 4G is not available, the system automatically switches to Wi-Fi. If both 4G and Wi-Fi are unavailable, the data is securely stored in the SD card for later upload.
- **Cloud Connectivity (MQTT):** Seamlessly connects to cloud platforms using the MQTT protocol for efficient data transmission and management.
- **Event Storage:** Offers on board event storage, capable of storing up to 50,000 events or more. Includes an external memory card slot for easy data retrieval in the event of 4G network issues.
- **Modbus RS485 Port:** Features 1 port of Modbus RS485 for integration with industrial control systems and sensors.
- **GPS Location Data:** Provides GPS location data for monitoring of remote equipment.
- **Micro Sim Compatibility:** Supports micro Sim card for flexible and convenient connectivity option.
- **Digital and Analog I/O Ports:** Multiple digital and analog input/output ports can be provided as per custom requirements.
- **Web-based UI Interface:** Intuitive web-based user interface for easy configuration, monitoring, and management.
- **Event Based Data reception:** Device can be programmed to receive data points of each event on digital inputs.
- **Dimensions:** 110 mm L x 80 mm B x 50 mm H.
- **Power:** Input Power – 15-40 VDC, Typical – 12V DC @ 150mA.
- **Operating Temperature:** 0 – 60C (32 ~ 140F).
- **Storage Temperature:** 20 - 70C (-4 ~ 158F).
- **Storage Humidity:** 5 ~ 95 % RH, non – Condensing.

Configuration Settings: -



- Power up the module by connecting a 12-24 V DC supply.
- Insert the SIM card into the designated slot.
- Place the Micro SD card into its designated slot.
- Connect MODBUS RTU Multiple Slave devices (up to 31) to the A & B Terminal Of Module.
- Access the module's configuration hotspot (ID: Wittelb-4G-GATEWAY, PASS: 123456789).
- Default IP Address of Module is "192.168.1.100/cpp1".
- Upon entering the provided IP address, a configuration webpage Pages will be accessible.

Page1:- TCP/IP Configuration:-

- Enter MAC ID of the Ethernet interface (6 hex bytes, e.g. 14:33:5C:77:62:48).
- Enter IP address for the Ethernet port (default: 192.168.1.254).
- Enter Subnet Mask (default: 255.255.255.0).
- Enter Gateway address (default: 192.168.1.1).
- Click Submit to save settings; the module applies to the new Ethernet configuration immediately.

Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	I/O Config	Monitor
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WiFi Configuration

SSID:

Password:

Device ID:

Data Frequency:

SD Data Upload Time:

Submit

TCP/IP Configuration

MAC ID: : : : : :

IP: . . .

MASK: . . .

GW: . . .

Submit

Page2:- On second page you can set device Baud Rate (9600 / 19200 / 38400 / 57600 / 115200), Data Bits (5 / 6 / 7 / 8), Parity (None / Odd / Even), and Stop Bits (1 / 2) for RS485 Communication

Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	I/O Config	logical oprations
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Rs485 Configuration

Baud Rate:

Data bits:

Parity:

Stop Bits:

Save RS485 Configuration

Page3:-

Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	cloud Platform	Modbus Master	I/O Config	logical oprations
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Cloud configuration

Host IP or Domain:

Port:

Publish Topic:

Publish Topic SD :

Subscribe Topic:

User Name:

Password:

Save Cloud Settings

- Navigate to page 3 on the cloud platform.
- Input the Host IP or Domain.
- Enter the port Number.
- Enter the topics for publishing and receiving messages.
- Provide a username and password if required.
- Allow a few seconds for the module to restart and establish a connection to MQTT.

Page4:-

- Go to page 4 on the Modbus Master interface.
- Input the name and slave address for the designated slave.
- Specify the register address from which you want to read slave data.
- Indicate the register end address for the slave.
- Enter the function code corresponding to the desired slave.

Wittelb 4G Gateway Configuration

 Network Configuration | Serial Port | Cloud Platform | **Modbus Master** | I/O Config | logical operations | About

Name	Slave Address	Register start	Quantity	Function Code	Actions
MFM	5	0	50	3	Delete

[Add Row](#)
[Submit](#)

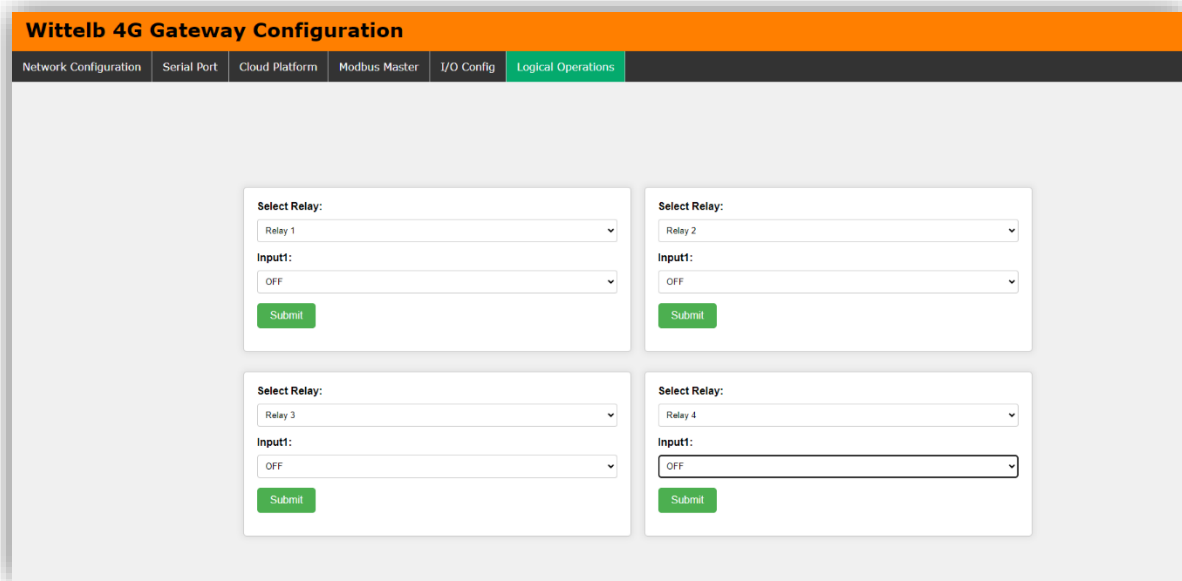
Page5:-

Wittelb 4G Gateway Configuration

 Network Configuration | Serial Port | Cloud Platform | Modbus Master | **I/O Config** | logical operations

Sl.No	Input/Output	Name	Value
1	DO1	RL11	
2	DO2	RL12	
3	DO3	RL13	1
4	DO4	RL14	
5	DI1	DI01	1
6	DI2	DI02	
7	DI3	DI03	
8	DI4	DI04	
9	AI1	AI-10-0	259
10	AI2	AI-10-1	197
11	AI3	AI-10-2	213
12	AI4	AI-10-3	227
13	AI1	AI-12-4	552
14	AI2	AI-12-5	2
15	AI3	AI-12-6	
16	AI4	AI-12-7	1

- Go to page 5 on the I/O Config interface.
- You can view real-time values of both analog and digital inputs on the baseboard.
- Additionally, you have the option to assign specific names to each input/output (I/O) as needed.



Wittelb 4G Gateway Configuration

Network Configuration | Serial Port | Cloud Platform | Modbus Master | I/O Config | **Logical Operations**

Select Relay:

Relay 1

Input1:

OFF

Submit

Select Relay:

Relay 2

Input1:

OFF

Submit

Select Relay:

Relay 3

Input1:

OFF

Submit

Select Relay:

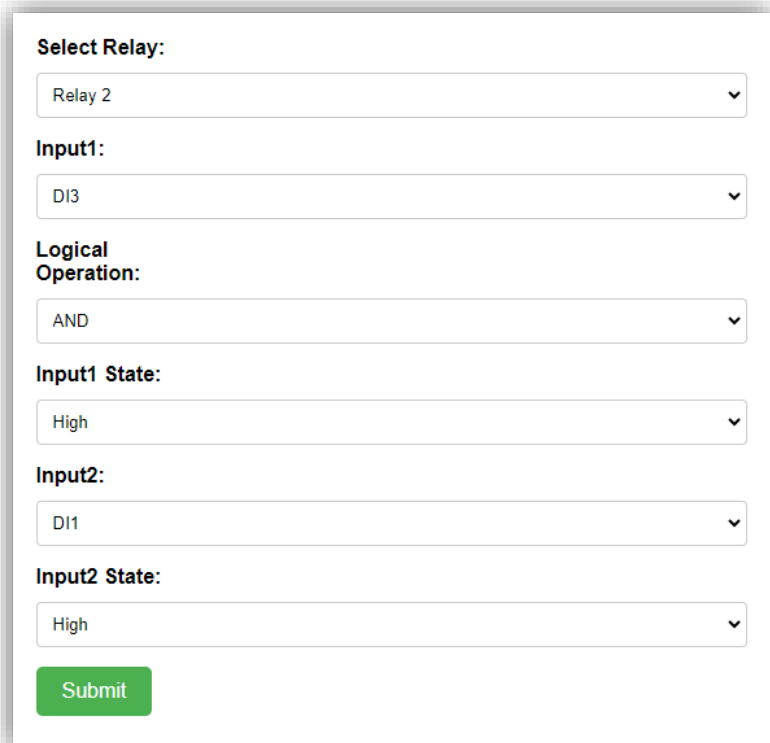
Relay 4

Input1:

OFF

Submit

- Navigate to page 6 on the Logical Operations interface.
- Here, you can configure logical AND and OR operations for each of the 4 relays.
- You have the flexibility to assign any two digital inputs to a relay and choose the desired logical operation.
- Additionally, you can assign an analog input to a relay and define minimum and maximum threshold values for relay operation.
- If you prefer not to use any logical operation, simply select the "OFF" option for input one.



Select Relay:

Relay 2

Input1:

DI3

Logical Operation:

AND

Input1 State:

High

Input2:

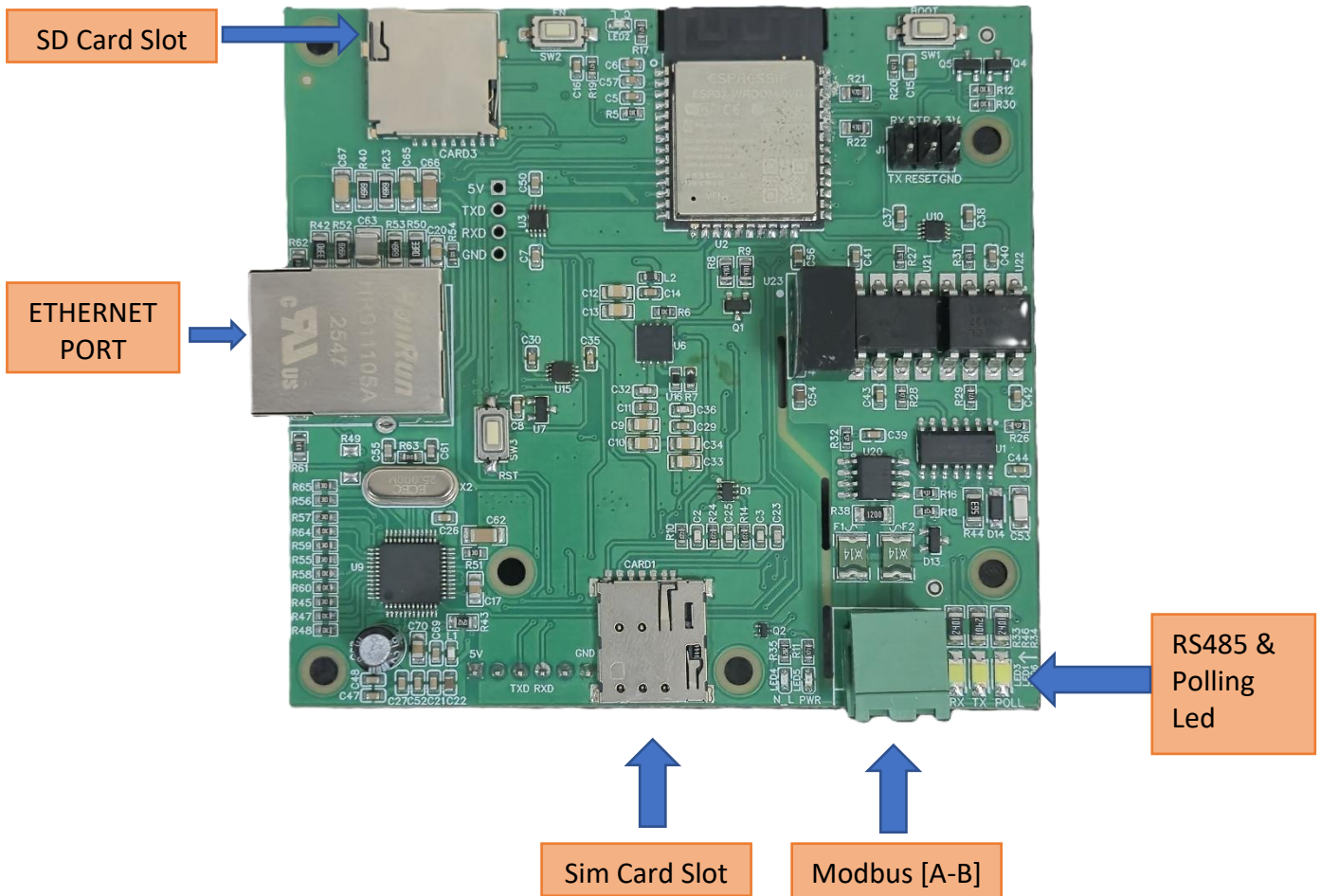
DI1

Input2 State:

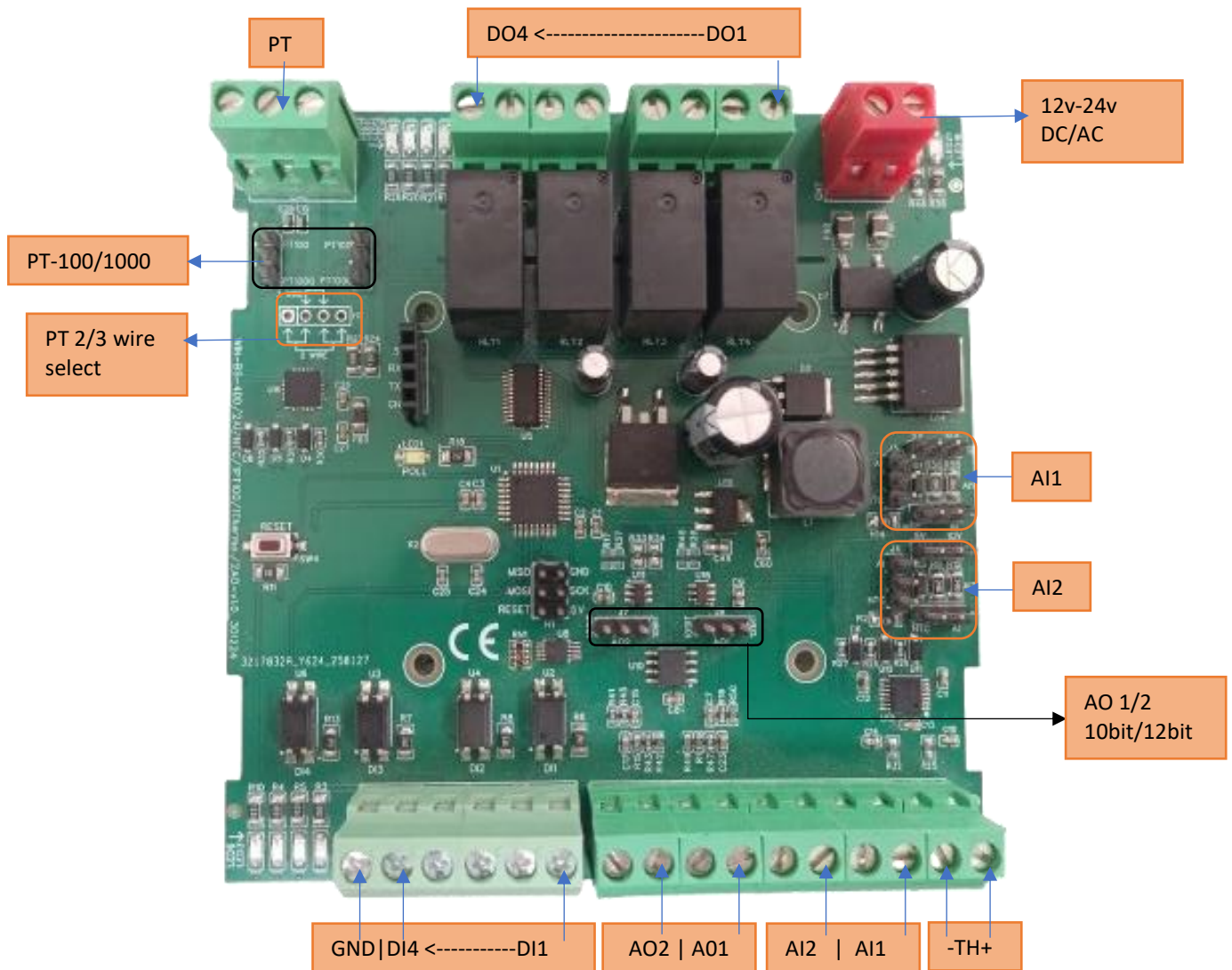
High

Submit

4G Gateway connection Diagram:



Base-Board connection Diagram:



PT100/1000 Thermocouple Module: -

Temperature Measurement System is a versatile and high-precision solution designed for diverse temperature sensing needs. With RTD (Pt100/Pt1000) measurement and for thermocouple measurement, this device is ideal for industrial, commercial, and scientific applications requiring precise temperature monitoring and remote data access.

1RTD Channel:

1. Features one channel for RTD measurements, capable of reading both Pt100 and Pt1000 sensors.
2. Utilizes advanced analog-to-digital conversion technology for accurate and reliable temperature readings.

1Thermocouple Channel:

1. Includes one channel for thermocouple measurements, compatible with various thermocouple types (**K, J, N, R, S, T, E, B**).
2. Ensures precise temperature monitoring across a wide range of applications.

Additional Features: -

Communication ports are isolated

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air

EFT IEC 61000-4-4, 50A (5/50ms)

400V isolation.

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